

6D SUGRA Bases — Multi-target External ($C^2 \in \{-5, \dots, -12\}$, Non-unimodular)

Total: 228 bases (after $\Delta_{\text{crit}} \leq 0$ filter).

Intersection matrix convention: Each matrix $(\cdot)_{b_0}$ is the intersection form of the SUGRA base, obtained by extending the LST intersection matrix by $b_0^2 = 9 - T_{\text{crit}}$ and then blowing down all (-1) -curves until no further blowdown is possible. The resulting matrix has signature $(1, T, n_0)$ where n_0 counts null directions (if any).

T	Count
2	5
3	10
4	17
5	18
6	24
7	28
8	28
9	48
10	50

$T = 2$ (5 bases)

Base: 11

$11 + 5$ [2-tgt] $\Delta = -267, |\det| = 4, (n_+, n_-, n_0) = (1, 2, 0)$

$T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 2, 1)$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -4 & -2 \\ 2 & -2 & 0 \end{pmatrix}_{b_0}$$

$11 + 6$ [2-tgt] $\Delta = -293, |\det| = 4, (n_+, n_-, n_0) = (1, 2, 0)$

$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 2, 1)$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -5 & -3 \\ 2 & -3 & -1 \end{pmatrix}_{b_0}$$

$11 + 7$ [2-tgt] $\Delta = -320, |\det| = 4, (n_+, n_-, n_0) = (1, 2, 0)$

$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 2, 1)$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -6 & -4 \\ 2 & -4 & -2 \end{pmatrix}_{b_0}$$

$11 + 8$ [2-tgt] $\Delta = -348, |\det| = 4, (n_+, n_-, n_0) = (1, 2, 0)$

$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 2, 1)$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -7 & -5 \\ 2 & -5 & -3 \end{pmatrix}_{b_0}$$

$11 + (12)$ [2-tgt] $\Delta = -463, |\det| = 4, (n_+, n_-, n_0) = (1, 2, 0)$

$T_{\text{crit}} = 17, (n_+, n_-, n_0)_{\text{crit}} = (1, 2, 1)$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -11 & -9 \\ 2 & -9 & -7 \end{pmatrix}_{b_0}$$

$T = 3$ (10 bases)

Base: 212

$^{5}_{212}$ $\Delta = -238, |\det| = 4, (n_+, n_-, n_0) = (1, 3, 0)$

$T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 3, 1)$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 1 \end{pmatrix}_{b_0}$$

Base: 121

$121 + 5$ [2-tgt] $\Delta = -238, |\det| = 4, (n_+, n_-, n_0) = (1, 3, 0)$

$T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 3, 1)$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 1 \end{pmatrix}_{b_0}$$

Base: 212

$^{6}_{212}$ $\Delta = -264, |\det| = 4, (n_+, n_-, n_0) = (1, 3, 0)$

$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 3, 1)$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -4 & -2 \\ 2 & -2 & 0 \end{pmatrix}_{b_0}$$

Base: 121

$121 + 6$ [2-tgt] $\Delta = -264, |\det| = 4, (n_+, n_-, n_0) = (1, 3, 0)$

$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 3, 1)$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -4 & -2 \\ 2 & -2 & 0 \end{pmatrix}_{b_0}$$

Base: 212

$^{7}_{212}$ $\Delta = -291, |\det| = 4, (n_+, n_-, n_0) = (1, 3, 0)$

$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 3, 1)$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -5 & -3 \\ 2 & -3 & -1 \end{pmatrix}_{b_0}$$

Base: 121

$121 + 7$ [2-tgt] $\Delta = -291, |\det| = 4, (n_+, n_-, n_0) = (1, 3, 0)$

$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 3, 1)$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -5 & -3 \\ 2 & -3 & -1 \end{pmatrix}_{b_0}$$

Base: 212

$$212^8 \quad \Delta = -319, |\det| = 4, (n_+, n_-, n_0) = (1, 3, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 3, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -6 & -4 \\ 2 & -4 & -2 \end{pmatrix}_{b_0}$$

Base: 121

$$121 + 8^{[2\text{-tgt}]} \quad \Delta = -319, |\det| = 4, (n_+, n_-, n_0) = (1, 3, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 3, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -6 & -4 \\ 2 & -4 & -2 \end{pmatrix}_{b_0}$$

Base: 212

$$2 \ 1 \ 2^{(12)} \quad \Delta = -434, |\det| = 4, (n_+, n_-, n_0) = (1, 3, 0)$$

$$T_{\text{crit}} = 17, (n_+, n_-, n_0)_{\text{crit}} = (1, 3, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -10 & -8 \\ 2 & -8 & -6 \end{pmatrix}_{b_0}$$

Base: 121

$$121 + (12)^{[2\text{-tgt}]} \quad \Delta = -434, |\det| = 4, (n_+, n_-, n_0) = (1, 3, 0)$$

$$T_{\text{crit}} = 17, (n_+, n_-, n_0)_{\text{crit}} = (1, 3, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -10 & -8 \\ 2 & -8 & -6 \end{pmatrix}_{b_0}$$

$T = 4$ **(17 bases)**

Base: 1221

$$1221 + 5^{[2\text{-tgt}]} \quad \Delta = -209, |\det| = 4, (n_+, n_-, n_0) = (1, 4, 0)$$

$$T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 4, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

Base: 131^1

$$131^1 + 5^{[2\text{-tgt}]} \quad \Delta = -217, |\det| = 4, (n_+, n_-, n_0) = (1, 4, 0)$$

$$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 4, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

Base: 122^2

$$221^2 5 \quad \Delta = -209, |\det| = 4, (n_+, n_-, n_0) = (1, 4, 0)$$

$$T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 4, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

Base: 1312

$$1312^5 \quad \Delta = -217, |\det| = 4, (n_+, n_-, n_0) = (1, 4, 0)$$

$$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 4, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 1 \end{pmatrix}_{b_0}$$

Base: 122^2

$$221^2 6 \quad \Delta = -235, |\det| = 4, (n_+, n_-, n_0) = (1, 4, 0)$$

$$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 4, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 1 \end{pmatrix}_{b_0}$$

Base: 1312

$$1312^6 \quad \Delta = -243, |\det| = 4, (n_+, n_-, n_0) = (1, 4, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 4, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -4 & -2 \\ 2 & -2 & 0 \end{pmatrix}_{b_0}$$

Base: 131^1

$$\begin{aligned} & \textcolor{blue}{131} + \textcolor{red}{6} \text{ [2-tgt]} \quad \Delta = -243, |\det| = 4, (n_+, n_-, n_0) = (1, 4, 0) \\ & T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 4, 1) \\ & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}_{b_0} \end{aligned}$$

Base: 1221

$$\begin{aligned} & \textcolor{blue}{1221} + \textcolor{red}{6} \text{ [2-tgt]} \quad \Delta = -235, |\det| = 4, (n_+, n_-, n_0) = (1, 4, 0) \\ & T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 4, 1) \\ & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 1 \end{pmatrix}_{b_0} \end{aligned}$$

Base: 2213

$$\begin{aligned} & \textcolor{red}{6} \text{ 2213} \quad \Delta = -243, |\det| = 9, (n_+, n_-, n_0) = (1, 4, 0) \\ & T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 4, 1) \\ & \begin{pmatrix} 0 & 3 & 2 \\ 3 & -3 & -1 \\ 2 & -1 & 0 \end{pmatrix}_{b_0} \end{aligned}$$

Base: $\textcolor{blue}{131}$

$$\begin{aligned} & \textcolor{blue}{131} + \textcolor{red}{6} \text{ [3-tgt]} \quad \Delta = -243, |\det| = 9, (n_+, n_-, n_0) = (1, 4, 0) \\ & T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 4, 1) \\ & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}_{b_0} \end{aligned}$$

Base: 1312

$$\begin{aligned} & \textcolor{blue}{1312} + \textcolor{red}{6} \text{ [2-tgt]} \quad \Delta = -243, |\det| = 9, (n_+, n_-, n_0) = (1, 4, 0) \\ & T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 4, 1) \\ & \begin{pmatrix} 0 & 3 & 2 \\ 3 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}_{b_0} \end{aligned}$$

Base: $\textcolor{red}{122}$

$$\begin{aligned} & \textcolor{red}{2217} \text{ } \quad \Delta = -262, |\det| = 4, (n_+, n_-, n_0) = (1, 4, 0) \\ & T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 4, 1) \\ & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -4 & -2 \\ 2 & -2 & 0 \end{pmatrix}_{b_0} \end{aligned}$$

Base: 1221

$$\begin{aligned} & \textcolor{blue}{1221} + \textcolor{red}{7} \text{ [2-tgt]} \quad \Delta = -262, |\det| = 4, (n_+, n_-, n_0) = (1, 4, 0) \\ & T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 4, 1) \\ & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -4 & -2 \\ 2 & -2 & 0 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} & \textcolor{blue}{1221} + \textcolor{red}{8} \text{ [2-tgt]} \quad \Delta = -290, |\det| = 4, (n_+, n_-, n_0) = (1, 4, 0) \\ & T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 4, 1) \\ & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -5 & -3 \\ 2 & -3 & -1 \end{pmatrix}_{b_0} \end{aligned}$$

Base: $\textcolor{red}{122}$

$$\begin{aligned} & \textcolor{red}{2218} \text{ } \quad \Delta = -290, |\det| = 4, (n_+, n_-, n_0) = (1, 4, 0) \\ & T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 4, 1) \\ & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -5 & -3 \\ 2 & -3 & -1 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} & \textcolor{red}{221(12)} \text{ } \quad \Delta = -405, |\det| = 4, (n_+, n_-, n_0) = (1, 4, 0) \\ & T_{\text{crit}} = 17, (n_+, n_-, n_0)_{\text{crit}} = (1, 4, 1) \\ & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -9 & -7 \\ 2 & -7 & -5 \end{pmatrix}_{b_0} \end{aligned}$$

Base: 1221

$$\begin{aligned} & \textcolor{blue}{1221} + \textcolor{red}{(12)} \text{ [2-tgt]} \quad \Delta = -405, |\det| = 4, (n_+, n_-, n_0) = (1, 4, 0) \\ & T_{\text{crit}} = 17, (n_+, n_-, n_0)_{\text{crit}} = (1, 4, 1) \\ & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -9 & -7 \\ 2 & -7 & -5 \end{pmatrix}_{b_0} \end{aligned}$$

$T = 5$ (18 bases)

Base: $12\bar{2}2$

$$2\bar{2}21\bar{5} \quad \Delta = -180, |\det| = 4, (n_+, n_-, n_0) = (1, 5, 0) \\ T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 5, 1) \\ \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: $2\bar{3}1\bar{3}$

$$\bar{5}1\bar{3}1\bar{3} \quad \Delta = -197, |\det| = 4, (n_+, n_-, n_0) = (1, 5, 0) \\ T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 5, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

Base: $1\bar{3}2\bar{1}$

$$1\bar{3}2\bar{1} + \bar{5} [2\text{-tgt}] \quad \Delta = -189, |\det| = 4, (n_+, n_-, n_0) = (1, 5, 0) \\ T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 5, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

Base: $1\bar{3}1\bar{3}1$

$$1\bar{3}1\bar{3}1 + \bar{5} [2\text{-tgt}] \quad \Delta = -196, |\det| = 4, (n_+, n_-, n_0) = (1, 5, 0) \\ T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 5, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

Base: $1\bar{3}2\bar{1}$

$$1\bar{3}2\bar{1}\bar{5} \quad \Delta = -189, |\det| = 4, (n_+, n_-, n_0) = (1, 5, 0) \\ T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 5, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

Base: $12\bar{2}2\bar{1}$

$$1\bar{2}2\bar{2}1 + \bar{5} [2\text{-tgt}] \quad \Delta = -180, |\det| = 4, (n_+, n_-, n_0) = (1, 5, 0) \\ T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 5, 1) \\ \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: $12\bar{2}2$

$$2\bar{2}21\bar{6} \quad \Delta = -206, |\det| = 4, (n_+, n_-, n_0) = (1, 5, 0) \\ T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 5, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

Base: $1\bar{3}1\bar{3}1$

$$1\bar{3}1\bar{3}1 + \bar{6} [2\text{-tgt}] \quad \Delta = -222, |\det| = 4, (n_+, n_-, n_0) = (1, 5, 0) \\ T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 5, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 1 \end{pmatrix}_{b_0}$$

Base: $1\bar{3}2\bar{1}$

$$1\bar{3}2\bar{1}\bar{6} \quad \Delta = -215, |\det| = 4, (n_+, n_-, n_0) = (1, 5, 0) \\ T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 5, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 1 \end{pmatrix}_{b_0}$$

Base: $12\bar{2}2\bar{1}$

$$1\bar{2}2\bar{2}1 + \bar{6} [2\text{-tgt}] \quad \Delta = -206, |\det| = 4, (n_+, n_-, n_0) = (1, 5, 0) \\ T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 5, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

Base: $12\bar{2}2$

$$2221\overset{2}{7} \quad \Delta = -233, |\det| = 4, (n_+, n_-, n_0) = (1, 5, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 5, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 1 \end{pmatrix}_{b_0}$$

Base: $1321\overset{2}$

$$1321\overset{2}{7} \quad \Delta = -242, |\det| = 4, (n_+, n_-, n_0) = (1, 5, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 5, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -4 & -2 \\ 2 & -2 & 0 \end{pmatrix}_{b_0}$$

Base: 12221

$$12221 + \overset{2}{7} \text{ [2-tgt]} \quad \Delta = -233, |\det| = 4, (n_+, n_-, n_0) = (1, 5, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 5, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$12221 + \overset{2}{8} \text{ [2-tgt]} \quad \Delta = -261, |\det| = 4, (n_+, n_-, n_0) = (1, 5, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 5, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -4 & -2 \\ 2 & -2 & 0 \end{pmatrix}_{b_0}$$

Base: $1222\overset{2}$

$$2221\overset{2}{8} \quad \Delta = -261, |\det| = 4, (n_+, n_-, n_0) = (1, 5, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 5, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -4 & -2 \\ 2 & -2 & 0 \end{pmatrix}_{b_0}$$

Base: $1321\overset{2}$

$$1321\overset{2}{8} \quad \Delta = -270, |\det| = 4, (n_+, n_-, n_0) = (1, 5, 0)$$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 5, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -5 & -3 \\ 2 & -3 & -1 \end{pmatrix}_{b_0}$$

Base: $1222\overset{2}$

$$2221(\overset{2}{12}) \quad \Delta = -376, |\det| = 4, (n_+, n_-, n_0) = (1, 5, 0)$$

$$T_{\text{crit}} = 17, (n_+, n_-, n_0)_{\text{crit}} = (1, 5, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -8 & -6 \\ 2 & -6 & -4 \end{pmatrix}_{b_0}$$

Base: 12221

$$12221 + (\overset{2}{12}) \text{ [2-tgt]} \quad \Delta = -376, |\det| = 4, (n_+, n_-, n_0) = (1, 5, 0)$$

$$T_{\text{crit}} = 17, (n_+, n_-, n_0)_{\text{crit}} = (1, 5, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -8 & -6 \\ 2 & -6 & -4 \end{pmatrix}_{b_0}$$

$T = 6$ **(24 bases)**

Base: $14132\overset{1}$

$$141\overset{51}{3}2 \quad \Delta = -188, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

Base: 122221

$$122221 + \overset{5}{5} \text{ [2-tgt]} \quad \Delta = -151, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0)$$

$$T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: $1232\overset{31}$

$$2\overset{31}{3}21\overset{5}{5} \quad \Delta = -170, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0)$$

$$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: 123131

$$123131 + \overset{5}{5} [2\text{-tgt}] \quad \Delta = -168, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0)$$

$$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: $2\overset{2}{3}131$

$$2\overset{2}{3}131\overset{5}{5} \quad \Delta = -170, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0)$$

$$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: $122\overset{2}{2}2$

$$2\overset{2}{2}221\overset{5}{5} \quad \Delta = -151, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0)$$

$$T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: $1\overset{1}{3}221$

$$1\overset{1}{3}221 + \overset{5}{5} [2\text{-tgt}] \quad \Delta = -160, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0)$$

$$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0}$$

Base: $12\overset{1}{3}21$

$$1\overset{1}{2}321 + \overset{5}{5} [2\text{-tgt}] \quad \Delta = -162, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0)$$

$$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0}$$

Base: 122221

$$122221 + \overset{6}{6} [2\text{-tgt}] \quad \Delta = -177, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0)$$

$$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: $1\overset{1}{2}321$

$$1\overset{1}{2}321 + \overset{6}{6} [2\text{-tgt}] \quad \Delta = -188, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

Base: $122\overset{2}{2}2$

$$2\overset{2}{2}221\overset{6}{6} \quad \Delta = -177, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0)$$

$$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: $12\overset{31}{3}2$

$$2\overset{31}{3}21\overset{6}{6} \quad \Delta = -196, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

Base: 123131

$$123131 + \overset{6}{6} [2\text{-tgt}] \quad \Delta = -194, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

Base: $2\bar{3}131$

$$2\bar{3}131\bar{6} \quad \Delta = -196, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0) \\ T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

Base: $12\bar{3}32$

$$2\bar{3}21\bar{7} \quad \Delta = -223, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0) \\ T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 1 \end{pmatrix}_{b_0}$$

Base: 122221

$$122221 + \bar{7} \text{ [2-tgt]} \quad \Delta = -204, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0) \\ T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

Base: $1222\bar{2}$

$$22221\bar{7} \quad \Delta = -204, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0) \\ T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

Base: $12\bar{3}21$

$$12\bar{3}21 + \bar{7} \text{ [2-tgt]} \quad \Delta = -215, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0) \\ T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

Base: $12\bar{3}32$

$$2\bar{3}21\bar{8} \quad \Delta = -251, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0) \\ T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & -4 & -2 \\ 2 & -2 & 0 \end{pmatrix}_{b_0}$$

Base: $12\bar{3}21$

$$12\bar{3}21 + \bar{8} \text{ [2-tgt]} \quad \Delta = -243, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0) \\ T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}_{b_0}$$

Base: $1222\bar{2}$

$$22221\bar{8} \quad \Delta = -232, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0) \\ T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 1 \end{pmatrix}_{b_0}$$

Base: 122221

$$122221 + \bar{8} \text{ [2-tgt]} \quad \Delta = -232, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0) \\ T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$122221 + (12) \text{ [2-tgt]} \quad \Delta = -347, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0) \\ T_{\text{crit}} = 17, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & -7 & -5 \\ 2 & -5 & -3 \end{pmatrix}_{b_0}$$

Base: $1222\bar{2}$

$$22221(12) \quad \Delta = -347, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0)$$

$$T_{\text{crit}} = 17, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -7 & -5 \\ 2 & -5 & -3 \end{pmatrix}_{b_0}$$

$T = 7$ (28 bases)

Base: $1 \overset{2}{131} 1$

$$1 \overset{5}{131} 1 \quad \Delta = -183, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

Base: 1314131

$$1314131 + \textcolor{red}{5} [2\text{-tgt}] \quad \Delta = -166, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: $141\overset{2}{321}$

$$141321\textcolor{red}{5} \quad \Delta = -161, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: 1223131

$$1223131 + \textcolor{red}{5} [2\text{-tgt}] \quad \Delta = -139, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: 1222221

$$1222221 + \textcolor{red}{5} [2\text{-tgt}] \quad \Delta = -122, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 5 \end{pmatrix}_{b_0}$$

Base: $1222\overset{2}{22}$

$$222221\textcolor{red}{5} \quad \Delta = -122, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 5 \end{pmatrix}_{b_0}$$

Base: $2151\overset{2}{31}$

$$215131\textcolor{red}{5} \quad \Delta = -183, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

Base: 1231321

$$1231321 + \textcolor{red}{5} [2\text{-tgt}] \quad \Delta = -140, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: $231\overset{2}{321}$

$$231321\textcolor{red}{5} \quad \Delta = -142, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: 1231321

$$1231321 + \textcolor{red}{6} [2\text{-tgt}] \quad \Delta = -166, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: 231321^2

$$2313216^2 \quad \Delta = -168, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: 122222^2

$$2222216^2 \quad \Delta = -148, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: 1222221

$$1222221 + 6^{[2\text{-tgt}]} \quad \Delta = -148, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: 1223131

$$1223131 + 6^{[2\text{-tgt}]} \quad \Delta = -165, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: 141321^2

$$1413216^2 \quad \Delta = -187, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

Base: 1314131

$$1314131 + 6^{[2\text{-tgt}]} \quad \Delta = -192, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

Base: 122222^2

$$2222217^2 \quad \Delta = -175, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: 141321^2

$$1413217^2 \quad \Delta = -214, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 1 \end{pmatrix}_{b_0}$$

Base: 231321^2

$$2313217^2 \quad \Delta = -195, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

Base: 1222221

$$1222221 + 7^{[2\text{-tgt}]} \quad \Delta = -175, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: 1231321

$$1231321 + 7 \text{ [2-tgt]} \quad \Delta = -193, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$\text{Base: } 141\overset{2}{3}21$$

$$1413218 \quad \Delta = -242, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 15, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -4 & -2 \\ 2 & -2 & 0 \end{pmatrix}_{b_0}$$

$$\text{Base: } 1222221$$

$$1222221 + 8 \text{ [2-tgt]} \quad \Delta = -203, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$\text{Base: } 231\overset{2}{3}21$$

$$2313218 \quad \Delta = -223, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$\text{Base: } 1231321$$

$$1231321 + 8 \text{ [2-tgt]} \quad \Delta = -221, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$\text{Base: } 1222\overset{2}{2}2$$

$$2222218 \quad \Delta = -203, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$222221(12) \quad \Delta = -318, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 17, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -6 & -4 \\ 2 & -4 & -2 \end{pmatrix}_{b_0}$$

$$\text{Base: } 1222221$$

$$1222221 + (12) \text{ [2-tgt]} \quad \Delta = -318, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0)$$

$$T_{\text{crit}} = 17, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -6 & -4 \\ 2 & -4 & -2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad (\mathbf{28 \text{ bases}})$$

$$\text{Base: } 13141\overset{2}{3}2$$

$$2 \overset{5}{3} 2 \quad \Delta = -140, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$\text{Base: } 13151\overset{1}{3}1$$

$$1315131 + 5 \text{ [2-tgt]} \quad \Delta = -161, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$\text{Base: } 12222221$$

$$12222221 + 5 \text{ [2-tgt]} \quad \Delta = -93, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0)$$

$$T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 6 \end{pmatrix}_{b_0}$$

Base: $31 \overset{131}{5} 13$

$$31 \overset{5131}{5} 13 \quad \Delta = -169, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: $1222222 \overset{2}{2}$

$$2222221 \overset{2}{5} \quad \Delta = -93, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0)$$

$$T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 6 \end{pmatrix}_{b_0}$$

Base: 12231321

$$12231321 + \overset{5}{2\text{-tgt}} \quad \Delta = -111, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0)$$

$$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 5 \end{pmatrix}_{b_0}$$

Base: 13141321

$$13141321 + \overset{5}{2\text{-tgt}} \quad \Delta = -138, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: $2313221 \overset{2}{2}$

$$2313221 \overset{2}{5} \quad \Delta = -113, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0)$$

$$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 5 \end{pmatrix}_{b_0}$$

Base: 13151231

$$13151231 + \overset{5}{2\text{-tgt}} \quad \Delta = -162, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: $31 \overset{131}{5} 13$

$$31 \overset{6131}{5} 13 \quad \Delta = -195, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0)$$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

Base: 12222221

$$12222221 + \overset{6}{2\text{-tgt}} \quad \Delta = -119, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0)$$

$$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 5 \end{pmatrix}_{b_0}$$

Base: 13141321

$$13141321 + \overset{6}{2\text{-tgt}} \quad \Delta = -164, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: $1314132 \overset{2}{2}$

$$2 \overset{613141}{3} 2 \quad \Delta = -166, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: $1315131 \overset{1}{1}$

$$1315131 + 6 \text{ [2-tgt]} \quad \Delta = -187, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0)$$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

Base: 1222222²

$$22222216 \quad \Delta = -119, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0)$$

$$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 5 \end{pmatrix}_{b_0}$$

Base: 2313221²

$$23132216 \quad \Delta = -139, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: 12231321

$$12231321 + 6 \text{ [2-tgt]} \quad \Delta = -137, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: 1222222²

$$22222217 \quad \Delta = -146, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: 12231321

$$12231321 + 7 \text{ [2-tgt]} \quad \Delta = -164, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: 12222221

$$12222221 + 7 \text{ [2-tgt]} \quad \Delta = -146, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: 2313221²

$$23132217 \quad \Delta = -166, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: 12222221

$$12222221 + 8 \text{ [2-tgt]} \quad \Delta = -174, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: 12231321

$$12231321 + 8 \text{ [2-tgt]} \quad \Delta = -192, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0)$$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

Base: 1222222²

$$22222218 \quad \Delta = -174, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: 2313221

2313221 $\Delta = -194$, $|\det| = 4$, $(n_+, n_-, n_0) = (1, 8, 0)$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

2313221 (12) $\Delta = -309$, $|\det| = 4$, $(n_+, n_-, n_0) = (1, 8, 0)$

$$T_{\text{crit}} = 18, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -6 & -4 \\ 2 & -4 & -2 \end{pmatrix}_{b_0}$$

Base: 12222221

12222221 + (12) [2-tgt] $\Delta = -289$, $|\det| = 4$, $(n_+, n_-, n_0) = (1, 8, 0)$

$$T_{\text{crit}} = 17, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -5 & -3 \\ 2 & -3 & -1 \end{pmatrix}_{b_0}$$

Base: 1222222

2222221 (12) $\Delta = -289$, $|\det| = 4$, $(n_+, n_-, n_0) = (1, 8, 0)$

$$T_{\text{crit}} = 17, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -5 & -3 \\ 2 & -3 & -1 \end{pmatrix}_{b_0}$$

$T = 9$ (48 bases)

Base: $31 \overset{1231}{5} 13$

$31 \overset{5}{5} 13$ $\Delta = -141$, $|\det| = 4$, $(n_+, n_-, n_0) = (1, 9, 0)$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: 131413221

131413221 + 5 [2-tgt] $\Delta = -109$, $|\det| = 4$, $(n_+, n_-, n_0) = (1, 9, 0)$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 5 \end{pmatrix}_{b_0}$$

Base: $131\epsilon_6 12231$

$131\epsilon_6 12231$ + 5 [2-tgt] $\Delta = -159$, $|\det| = 4$, $(n_+, n_-, n_0) = (1, 9, 0)$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: 13151321

13151321 + 5 [2-tgt] $\Delta = -133$, $|\det| = 4$, $(n_+, n_-, n_0) = (1, 9, 0)$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}_{b_0}$$

Base: 122313221

122313221 + 5 [2-tgt] $\Delta = -82$, $|\det| = 4$, $(n_+, n_-, n_0) = (1, 9, 0)$

$$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 6 \end{pmatrix}_{b_0}$$

Base: 123215131

123215131 + 5 [2-tgt] $\Delta = -135$, $|\det| = 4$, $(n_+, n_-, n_0) = (1, 9, 0)$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: $1 \overset{1321}{\epsilon_6} 131$

$1 \overset{1231}{\epsilon_6} 131$ + 5 [2-tgt] $\Delta = -159$, $|\det| = 4$, $(n_+, n_-, n_0) = (1, 9, 0)$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: $131\overset{1}{\underset{31}{\epsilon_6}}13$

$$31\overset{1}{\underset{31}{\epsilon_6}}131\textcolor{red}{5} \quad \Delta = -166, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: $1321\epsilon_61231$

$$1321\epsilon_61231 + \textcolor{red}{5} [2\text{-tgt}] \quad \Delta = -160, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: $1\overset{1}{\underset{131}{\epsilon_6}}131$

$$1\overset{1}{\underset{131}{\epsilon_6}}131 + \textcolor{red}{5} [2\text{-tgt}] \quad \Delta = -158, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0}$$

Base: 122222221

$$122222221 + \textcolor{red}{5} [2\text{-tgt}] \quad \Delta = -64, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 7 \end{pmatrix}_{b_0}$$

Base: 123151231

$$123151231 + \textcolor{red}{5} [2\text{-tgt}] \quad \Delta = -134, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: 131414131

$$131414131 + \textcolor{red}{5} [2\text{-tgt}] \quad \Delta = -136, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: $123141\overset{2}{3}2$

$$\overset{5}{123141} \begin{matrix} 2 & 3 & 2 \end{matrix} \quad \Delta = -112, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 5 \end{pmatrix}_{b_0}$$

Base: $1321\overset{31}{\underset{31}{\epsilon_6}}13$

$$31\overset{31}{\underset{31}{\epsilon_6}}1231\textcolor{red}{5} \quad \Delta = -167, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: $122222\overset{2}{3}2$

$$22222221\textcolor{red}{5} \quad \Delta = -64, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 7 \end{pmatrix}_{b_0}$$

Base: $131\overset{31}{\underset{31}{\epsilon_6}}123$

$$321\overset{31}{\underset{31}{\epsilon_6}}131\textcolor{red}{5} \quad \Delta = -167, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: $21 \overset{131}{\mathfrak{e}_6} 131$

$$21 \overset{131}{\mathfrak{e}_6} 131 + \textcolor{blue}{5} \text{ [2-tgt]} \quad \Delta = -158, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: 123141321

$$123141321 + \textcolor{blue}{5} \text{ [2-tgt]} \quad \Delta = -110, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 5 \end{pmatrix}_{b_0}$$

Base: $31 \overset{1231}{5} 13$

$$31 \overset{61231}{5} 13 \quad \Delta = -167, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: 131413221

$$131413221 + \textcolor{red}{6} \text{ [2-tgt]} \quad \Delta = -135, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: $12222222 \overset{2}{2}$

$$22222221 \overset{2}{\textcolor{red}{6}} \quad \Delta = -90, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 6 \end{pmatrix}_{b_0}$$

Base: $131 \overset{31}{\mathfrak{e}_6} 123$

$$321 \overset{31}{\mathfrak{e}_6} 131 \textcolor{red}{6} \quad \Delta = -193, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 15, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

Base: $12314132 \overset{2}{2}$

$$2 \overset{6123141}{3} 2 \quad \Delta = -138, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: $1 \overset{1}{\mathfrak{e}_6} 131 \overset{131}{131}$

$$1 \overset{1}{\mathfrak{e}_6} 131 + \textcolor{red}{6} \text{ [2-tgt]} \quad \Delta = -184, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 15, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

Base: $131 \overset{1}{\mathfrak{e}_6} 13 \overset{31}{31}$

$$31 \overset{1}{\mathfrak{e}_6} 131 \textcolor{red}{6} \quad \Delta = -192, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 15, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

Base: 122222221

$$122222221 + \textcolor{red}{6} \text{ [2-tgt]} \quad \Delta = -90, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 6 \end{pmatrix}_{b_0}$$

Base: 123215131

$$123215131 + 6 \text{ [2-tgt]} \quad \Delta = -161, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 14, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: 123141321

$$123141321 + 6 \text{ [2-tgt]} \quad \Delta = -136, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 13, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: $21 \overset{131}{\mathfrak{t}_6} 131$

$$21 \overset{131}{\mathfrak{t}_6} 131 + 6 \text{ [2-tgt]} \quad \Delta = -184, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 15, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

Base: 122313221

$$122313221 + 6 \text{ [2-tgt]} \quad \Delta = -108, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 12, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 5 \end{pmatrix}_{b_0}$$

Base: $13151321 \overset{1}{\mathfrak{t}_6}$

$$13151321 + 6 \text{ [2-tgt]} \quad \Delta = -159, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 14, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0}$$

Base: 131414131

$$131414131 + 6 \text{ [2-tgt]} \quad \Delta = -162, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 14, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: 122313221

$$122313221 + 7 \text{ [2-tgt]} \quad \Delta = -135, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 13, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: 122222221

$$122222221 + 7 \text{ [2-tgt]} \quad \Delta = -117, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 12, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 5 \end{pmatrix}_{b_0}$$

Base: 123141321

$$123141321 + 7 \text{ [2-tgt]} \quad \Delta = -163, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 14, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: $12314132 \overset{2}{\mathfrak{t}_6}$

$$7 \overset{123141}{\mathfrak{t}_6} 123141 \quad \Delta = -165, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 14, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: $31 \overset{1231}{\mathfrak{t}_5} 13$

$$7 \overset{1231}{\mathfrak{t}_5} 1231 \quad \Delta = -194, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 15, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

Base: $12222222 \overset{2}{\mathfrak{t}_6}$

$$22222221\overset{2}{7} \quad \Delta = -117, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$\text{Base: } 12314132\overset{2}{2}$$

$$2\overset{8}{123141}3\overset{2}{2} \quad \Delta = -193, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 15, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$\text{Base: } 122222221$$

$$122222221 + \overset{8}{2} \text{ [2-tgt]} \quad \Delta = -145, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$\text{Base: } 12222222\overset{2}{2}$$

$$22222221\overset{2}{8} \quad \Delta = -145, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$\text{Base: } 122313221$$

$$122313221 + \overset{8}{2} \text{ [2-tgt]} \quad \Delta = -163, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

$$\text{Base: } 31\overset{1231}{5}13$$

$$31\overset{8}{1231}5\overset{13}{13} \quad \Delta = -222, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 16, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$\text{Base: } 123141321$$

$$123141321 + \overset{8}{2} \text{ [2-tgt]} \quad \Delta = -191, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 15, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$\text{Base: } 122313221$$

$$122313221 + (\textcolor{red}{12}) \text{ [2-tgt]} \quad \Delta = -278, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 18, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -5 & -3 \\ 2 & -3 & -1 \end{pmatrix}_{b_0}$$

$$\text{Base: } 122222221$$

$$122222221 + (\textcolor{red}{12}) \text{ [2-tgt]} \quad \Delta = -260, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 17, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -4 & -2 \\ 2 & -2 & 0 \end{pmatrix}_{b_0}$$

$$\text{Base: } 12222222\overset{2}{2}$$

$$22222221(\textcolor{red}{12}) \quad \Delta = -260, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0)$$

$$T_{\text{crit}} = 17, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -4 & -2 \\ 2 & -2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad (\text{50 bases})$$

$$\text{Base: } 12222222\overset{2}{2}$$

$$222222221\overset{2}{5} \quad \Delta = -35, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}_{b_0}$$

$$\text{Base: } 131\overset{1}{5}13221$$

$$131\overset{1}{5}13221 + \overset{1}{5} [2\text{-tgt}] \quad \Delta = -104, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$\text{Base: } 1223151231$$

$$1223151231 + \overset{1}{5} [2\text{-tgt}] \quad \Delta = -105, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$\text{Base: } 1231413221$$

$$1231413221 + \overset{1}{5} [2\text{-tgt}] \quad \Delta = -81, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 6 \end{pmatrix}_{b_0}$$

$$\text{Base: } 1\overset{1321}{\mathfrak{e}_7'} 1231$$

$$1\overset{1231}{\mathfrak{e}_7'} 1231 + \overset{1}{5} [2\text{-tgt}] \quad \Delta = -158, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 15, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

$$\text{Base: } 1314141\overset{2}{32}$$

$$\overset{5}{1314141} 3 \quad \Delta = -110, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$\text{Base: } 1223141\overset{2}{32}$$

$$\overset{5}{1223141} 3 \quad \Delta = -83, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 6 \end{pmatrix}_{b_0}$$

$$\text{Base: } 1222222221$$

$$1222222221 + \overset{1}{5} [2\text{-tgt}] \quad \Delta = -35, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \end{pmatrix}_{b_0}$$

$$\text{Base: } 12321\overset{31}{\mathfrak{e}_6} 13$$

$$31\overset{31}{\mathfrak{e}_6} 12321\overset{1}{5} \quad \Delta = -140, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$\text{Base: } 1231512321$$

$$1231512321 + \overset{1}{5} [2\text{-tgt}] \quad \Delta = -107, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$\text{Base: } 1321\mathfrak{e}_6 12321$$

$$1321\mathfrak{e}_6 12321 + \overset{1}{5} [2\text{-tgt}] \quad \Delta = -133, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: $1 \overset{12321}{\epsilon_6} 131$

$$1 \overset{12321}{\epsilon_6} 131 + \textcolor{red}{5} [2\text{-tgt}] \quad \Delta = -132, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: $1321\epsilon'_7 12231$

$$1321\epsilon'_7 12231 + \textcolor{red}{5} [2\text{-tgt}] \quad \Delta = -158, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 15, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: 1314141321

$$1314141321 + \textcolor{red}{5} [2\text{-tgt}] \quad \Delta = -108, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 5 \end{pmatrix}_{b_0}$$

Base: $31 \overset{12231}{5} 13$

$$31 \overset{5}{12231} 13 \quad \Delta = -112, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 5 \end{pmatrix}_{b_0}$$

Base: $4 \overset{22}{\otimes} \overset{31}{\epsilon_6} 131$

$$4 \overset{22}{\otimes} \overset{31}{\epsilon_6} 131 \textcolor{red}{5} \quad \Delta = -165, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 15, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: 123151321

$$123151321 + \textcolor{red}{5} [2\text{-tgt}] \quad \Delta = -105, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

Base: 1231413221

$$1231413221 + \textcolor{red}{6} [2\text{-tgt}] \quad \Delta = -107, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 5 \end{pmatrix}_{b_0}$$

Base: $1 \overset{12321}{\epsilon_6} 131$

$$1 \overset{12321}{\epsilon_6} 131 + \textcolor{red}{6} [2\text{-tgt}] \quad \Delta = -158, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 15, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: 1222222221

$$1222222221 + \textcolor{red}{6} [2\text{-tgt}] \quad \Delta = -61, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 7 \end{pmatrix}_{b_0}$$

Base: $1314141\overset{2}{3}2$

$$\overset{6}{1314141} 3 \overset{2}{2} \quad \Delta = -136, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: $1222222\overset{2}{2}2$

$$222222221\overset{2}{6} \quad \Delta = -61, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 5 \\ 2 & 5 & 7 \end{pmatrix}_{b_0}$$

Base: 1314141321

$$1314141321 + \overset{22}{6} [2\text{-tgt}] \quad \Delta = -134, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: $4 \otimes \overset{22}{\mathfrak{e}_6} \overset{31}{131}$

$$4 \otimes \overset{22}{\mathfrak{e}_6} \overset{31}{131}\overset{22}{6} \quad \Delta = -191, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 16, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

Base: 1231512321

$$1231512321 + \overset{22}{6} [2\text{-tgt}] \quad \Delta = -133, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: 131513221^1

$$131513221 + \overset{1}{6} [2\text{-tgt}] \quad \Delta = -130, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}_{b_0}$$

Base: $31 \overset{12231}{5} 13$

$$31 \overset{6}{5} \overset{12231}{13} \quad \Delta = -138, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: $12321\overset{31}{\mathfrak{e}_6} 13$

$$31\overset{31}{\mathfrak{e}_6} 12321\overset{31}{6} \quad \Delta = -166, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 15, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: $1223141\overset{2}{3}2$

$$\overset{6}{2} \overset{1223141}{3} \overset{2}{2} \quad \Delta = -109, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 5 \end{pmatrix}_{b_0}$$

Base: 123151321^1

$$123151321 + \overset{1}{6} [2\text{-tgt}] \quad \Delta = -131, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}_{b_0}$$

Base: 1231413221

$$1231413221 + \overset{7}{7} [2\text{-tgt}] \quad \Delta = -134, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: 1231512321

$$1231512321 + \overset{7}{7} [2\text{-tgt}] \quad \Delta = -160, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 15, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: 12321³¹ $\mathfrak{e}_6$ 13

$$31\mathfrak{e}_6^{31}12321\textcolor{red}{7} \quad \Delta = -193, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 16, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0}$$

Base: 122222²22

$$22222221\textcolor{red}{7} \quad \Delta = -88, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 6 \end{pmatrix}_{b_0}$$

Base: 12315¹1321

$$12315\textcolor{blue}{1}321 + \textcolor{red}{7} \text{ [2-tgt]} \quad \Delta = -158, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 15, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0}$$

Base: 122222221

$$122222221 + \textcolor{red}{7} \text{ [2-tgt]} \quad \Delta = -88, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 12, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 4 \\ 2 & 4 & 6 \end{pmatrix}_{b_0}$$

Base: 1223141²32

$$2\textcolor{red}{7}^{1223141}3^2 \quad \Delta = -136, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 14, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 4 \end{pmatrix}_{b_0}$$

Base: 31¹²²³¹5¹³

$$31\textcolor{red}{7}^{12231}5^{13} \quad \Delta = -165, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 15, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: 12321³¹ $\mathfrak{e}_6$ 13

$$31\mathfrak{e}_6^{31}12321\textcolor{red}{8} \quad \Delta = -221, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 17, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 1 \end{pmatrix}_{b_0}$$

Base: 1231413221

$$1231413221 + \textcolor{red}{8} \text{ [2-tgt]} \quad \Delta = -162, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 15, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: 122222²22

$$22222221\textcolor{red}{8} \quad \Delta = -116, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 13, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 5 \end{pmatrix}_{b_0}$$

Base: 1223141²32

$$2\textcolor{red}{8}^{1223141}3^2 \quad \Delta = -164, |\det| = 4, (n_+, n_-, n_0) = (1, 10, 0)$$

$$T_{\text{crit}} = 15, (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1)$$

$$\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}_{b_0}$$

Base: 12315¹1321

$$\begin{aligned} & \textcolor{blue}{123151321} + \textcolor{red}{8}^{[2\text{-tgt}]} \quad \Delta = -186, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 10, 0) \\ & T_{\text{crit}} = 16, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1) \\ & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \end{aligned}$$

Base: 1231512321

$$\begin{aligned} & \textcolor{blue}{1231512321} + \textcolor{red}{8}^{[2\text{-tgt}]} \quad \Delta = -188, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 10, 0) \\ & T_{\text{crit}} = 16, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1) \\ & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0} \end{aligned}$$

Base: $31 \overset{12231}{5} 13$

$$\begin{aligned} & 31 \overset{\textcolor{red}{8}^{12231}}{5} 13 \quad \Delta = -193, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 10, 0) \\ & T_{\text{crit}} = 16, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1) \\ & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -2 & 0 \\ 2 & 0 & 2 \end{pmatrix}_{b_0} \end{aligned}$$

Base: 1222222221

$$\begin{aligned} & \textcolor{blue}{1222222221} + \textcolor{red}{8}^{[2\text{-tgt}]} \quad \Delta = -116, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 10, 0) \\ & T_{\text{crit}} = 13, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1) \\ & \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 5 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} & \textcolor{blue}{1222222221} + \textcolor{red}{(12)}^{[2\text{-tgt}]} \quad \Delta = -231, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 10, 0) \\ & T_{\text{crit}} = 17, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1) \\ & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 1 \end{pmatrix}_{b_0} \end{aligned}$$

Base: $1223141\overset{2}{3}2$

$$\begin{aligned} & \textcolor{red}{(12)}^{1223141} 2 \quad 3 \quad 2 \quad \Delta = -279, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 10, 0) \\ & T_{\text{crit}} = 19, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1) \\ & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -5 & -3 \\ 2 & -3 & -1 \end{pmatrix}_{b_0} \end{aligned}$$

Base: $31 \overset{12231}{5} 13$

$$\begin{aligned} & \textcolor{red}{(12)}^{12231} 31 \quad 5 \quad 13 \quad \Delta = -308, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 10, 0) \\ & T_{\text{crit}} = 20, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1) \\ & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -6 & -4 \\ 2 & -4 & -2 \end{pmatrix}_{b_0} \end{aligned}$$

Base: $1222222\overset{2}{2}2$

$$\begin{aligned} & \overset{2}{222222221} \textcolor{red}{(12)} \quad \Delta = -231, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 10, 0) \\ & T_{\text{crit}} = 17, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1) \\ & \begin{pmatrix} 0 & 2 & 2 \\ 2 & -3 & -1 \\ 2 & -1 & 1 \end{pmatrix}_{b_0} \end{aligned}$$